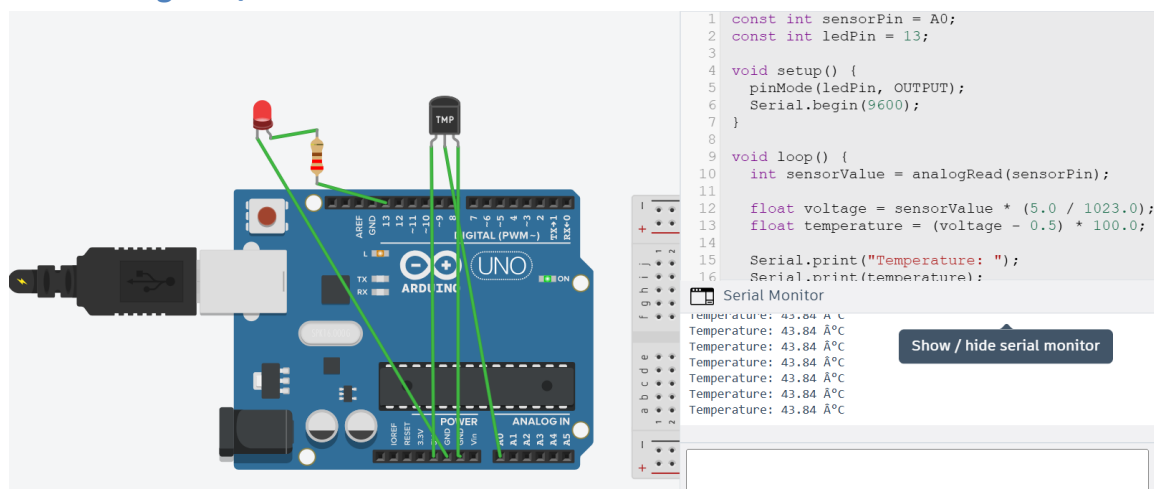




Circuit Connections

TMP36 Left Pin → 5V

TMP36 Middle Pin → A0

TMP36 Right Pin → GND

LED Anode (+) → D13 through 220 Ω resistor

LED Cathode (-) → GND

Working Principle

The TMP36 senses the machine temperature and produces an analog voltage. Arduino reads this value through analog pin A0, converts it into temperature in degrees Celsius, and displays the result on the Serial Monitor. When the temperature rises above 35°C, Arduino switches ON the warning LED. When the temperature falls below the threshold, the LED switches OFF.

Arduino Code Explanation

sensorPin stores the analog input pin (A0).

ledPin stores the LED output pin (D13).

setup() initializes the LED pin and Serial Monitor.

loop() repeatedly reads the sensor, converts the voltage into temperature, prints the value, compares it with 35°C, and controls the LED.

Observed Output

During simulation the measured temperature was approximately 43.84°C. Since this value is greater than 35°C, the warning LED turned ON, indicating an overheating condition.

Advantages

- Low-cost monitoring system
- Easy to implement
- Improves machine safety
- Supports preventive maintenance
- Can be expanded with IoT cloud monitoring

Applications

Industrial automation

Factory machine monitoring

Predictive maintenance

Boiler and motor monitoring

Industry 4.0 laboratories

Future Scope

The project can be upgraded by adding Wi-Fi (ESP8266/ESP32), cloud dashboards, SMS/email alerts, data logging, LCD display, and multiple temperature sensors.

Conclusion

This project successfully demonstrates a basic smart factory monitoring system using Arduino and a TMP36 sensor. It provides continuous temperature monitoring and an overheating indication, making it suitable for learning industrial IoT concepts.